Subject-Disjoint Benchmarking of Lightweight Detectors for Visible Driver-Cue Localization

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***Abstract*—Road traffic injuries remain a leading cause of death worldwide, and driver monitoring systems (DMSs) are increasingly mandated by vehicle-safety assessment programmes. Object detection can spatially localize visible evidence of driver inattention inside a vehicle cabin, yet existing evaluations often lack subject-disjoint splits, retain no negative frames, or select operating points on test data. We present a reproducible benchmark that addresses these gaps. The primary red–green–blue (RGB) experiment retains 15,723 Driver Monitoring Dataset frames from 14 drivers, including 12,722 negatives, and evaluates three nano-scale detection systems — You Only Look Once (YOLO) 11n, YOLO26n, and nano Fine-grained Distribution Refinement (D-FINE-N) — across three paired seeds under a strict validation-only threshold protocol. No single detector dominates: YOLO11n leads coarse-IoU mean average precision (mAP$_{50}$) at $91.79 \pm 0.44\%$ and throughput at 68.0 ± 1.3 frames per second (FPS), while D-FINE-N leads micro-F_1 at $91.14 \pm 2.66\%$ but runs slowest at 40.4 ± 1.2 FPS. Between-subject variation exceeds seed variation, and class imbalance materially affects pooled metrics. In the completed eight-system near-infrared (NIR) exposure study, 1:6 negative exposure produces no uniform strict-IoU gain; RT-DETRv2-S leads strict-IoU AP, whereas YOLOv8n leads micro-F_1 at both exposures. Claims concern visible-cue localization, not temporal fatigue or behavior inference.**

***Index Terms*—driver monitoring, object detection, subject-disjoint evaluation, visible driver cues, near-infrared imaging, real-time inference**

I. Introduction

Road traffic injuries claim approximately 1.19 million lives annually and impose a substantial economic burden worldwide [1]. In response, Euro NCAP now awards safety-assist credit for camera-based driver monitoring [2], accelerating the deployment of in-cabin driver monitoring systems (DMSs). These systems must operate robustly under changes in driver identity, illumination, viewpoint, and occlusion. Whole-frame recognition can describe an activity, but it may also exploit identity or cabin context without showing the visual evidence used for a decision. Object detection provides a complementary formulation: it localizes cue-specific evidence, makes false positives and misses spatially interpretable, and measures detections on frames containing no target cue. A single-frame detector also avoids the latency and architectural complexity of a temporal model when real-time throughput is the experimental focus. This formulation does *not* infer fatigue, intent, duration, or another temporal driver state.

DMD provides multimodal attention and alertness recordings [3], while Drive&Act contains synchronized color, NIR, depth, and pose streams for fine-grained activities [4]. Both contain temporally correlated video and repeated observations of each driver. Evaluation can therefore be optimistic if neighboring frames or identities cross partitions, natural negative frames are removed, the test set determines the confidence threshold, or latency boundaries differ. Existing driver-detection studies on DMD [5] and posture-classification benchmarks such as the State Farm challenge [6] further illustrate the diversity of evaluation choices and the difficulty of cross-paper comparison.

Recent work proposes specialized lightweight networks for in-cabin recognition and detection [7]–[10]. Our objective is different: to compare complete, publicly released nano detector systems without task-specific architectural modification. The contributions are:

- a primary subject-disjoint RGB benchmark for four explicitly defined visible cues, with all sampled frames preserved and annotation structure automatically audited;
- validation-only operating-point selection followed by protected testing, three paired RGB seeds, class- and subject-level analysis, false detections on negative frames, and synchronized latency; and
- a checksum-backed reproduction workflow plus a clearly separated, one-seed, eight-system NIR training-negative exposure study with deterministic 1-FPS midpoint sampling.

II. Related Work

A. Driver Recognition and Cue Localization

MobileVGG and convolution–transformer hybrids demonstrate efficient distracted-driver classification [7], [11]. PO-GUISE+ uses pose and object guidance for efficient driver-action recognition in Drive&Act NIR imagery [12]. Such whole-frame or temporal recognition tasks are not interchangeable with the present bounding-box task. Here, a correct prediction must identify both a cue class and its spatial evidence in one frame; no temporal state is estimated.

Detection-specific work modifies YOLO backbones, attention, or multiscale aggregation for dangerous-driving or fatigue cues [8]–[10]. Comparative studies assess convolutional neural networks (CNNs) against transformers [13] or several YOLO generations [14]. The YOLO family originates with single-shot grid regression [15] and has evolved through multiple architectural revisions [16]–[18]. However, differences in driver separation, retained negatives, operating-point calibration, and runtime protocol limit direct cross-paper ranking.

B. Released Detector Systems

The Detection Transformer (DETR) casts object detection as bipartite set prediction [19]; the Real-Time Detection Transformer (RT-DETR) makes this formulation practical for real-time use [20]; and D-FINE refines box-edge distributions for improved localization [21]. The RGB and NIR comparisons use D-FINE-N, YOLO11n [17], and YOLO26n [18] as shipped systems. Native training and inference recipes remain part of each system: the pinned YOLO26 and D-FINE paths are end-to-end, whereas YOLO11n retains non-maximum suppression (NMS) [22]. Recent edge-deployment work demonstrates that such lightweight systems can approach real-time performance on low-cost hardware [23]. This is therefore a system-level benchmark, not an isolation of architecture alone.

III. Methodology

A. Task Definition and Workflow

The task is bounding-box localization of four visible RGB cues: yawning, hand over mouth, drinking, and phone use. Detection is used because it spatially localizes the evidence, discourages whole-frame identity and cabin shortcuts, and exposes errors on negative frames. Figure 1 summarizes the shared protocol. Dataset construction, splitting, training, and validation precede a checksum and confirmation gate; no test-derived threshold or model change is permitted. The RGB benchmark is primary. NIR is a distinct secondary experiment rather than a paired modality comparison.

B. RGB Dataset and Annotation Protocol

The RGB dataset contains 15,723 face crops sampled at 1 FPS from 81 DMD recordings of 14 drivers. One primary annotator labeled the data in a single smooth pass; no independent second-person review was established. Every sampled frame was preserved to avoid post-sampling curation, yielding 3,001 positive and 12,722 negative frames. Table I freezes the spatial ontology. Box semantics intentionally differ because the visible evidence differs by cue.

Each frame contains zero or one box. Yawning and hand-over-mouth are mutually exclusive; when a hand covers a yawn, hand-over-mouth takes precedence. Boxes enclose only the visible target. Occluded, truncated, or out-of-frame anatomy is never estimated, although a partially occluded cue remains valid when visually identifiable. No arbitrary minimum-pixel cutoff is applied at 640×640.

TABLE I
Frozen RGB Box Semantics and Test Support

Cue	Bounding-box target	Test n	Median area
Yawning	Mouth	33	1.38%
Hand over mouth	Full face and covering hand	28	24.67%
Drinking	Visible hand/container–mouth interaction	56	16.09%
Phone use	Visible hand–phone interaction	497	8.37%

TABLE II
Shared RGB Training Configuration

Setting	Value
Input resolution	640×640
Physical / effective batch	8 / 32 (4-step accumulation)
Epochs	220 (no early stopping)
Precision	AMP (FP16/FP32)
Pretrained backbone	COCO
Seeds	$\{13, 37, 73\}$ (predeclared)
YOLO optimizer	`auto` + linear schedule
D-FINE optimizer	AdamW + MultiStepLR

An automated COCO audit found 15,723 unique images and 3,001 unique annotations, with zero duplicate identifiers or file names, orphan annotations, invalid categories, nonfinite or nonpositive boxes, out-of-bounds boxes, or multi-annotation frames. The smallest annotated area was 0.81% of the frame; 139 of 159 yawning boxes occupied below 2%. This is a structural audit, not evidence of inter-annotator agreement or semantic correctness.

C. Split, Models, and Training

An exhaustive 8:3:3 subject assignment minimizes the worst relative class-distribution deviation, with root-mean-square deviation as a tie-breaker. The frozen training, validation, and test partitions contain 9,087, 3,423, and 3,213 frames. The three test subjects contribute 1,080, 920, and 1,213 frames and 178, 183, and 253 positives, respectively.

COCO-pretrained YOLO11n [17], YOLO26n [18], and D-FINE-N [21] use 640×640 input, physical batch size 8, four-step gradient accumulation for an effective batch of 32, 220 epochs, automatic mixed precision (AMP), and no early stopping. A fixed patch normalizes an incomplete final accumulation window by its actual sample count. Native recipes are retained: Ultralytics `optimizer=auto` with a linear schedule, and the official D-FINE AdamW/MultiStepLR recipe. Seeds $\{13, 37, 73\}$ were declared in advance, and no run was selected for presentation. Table II consolidates the shared settings.

D. Validation, Test Metrics, and Profiling

The best model-native validation checkpoint is retained. Its confidence threshold τ^* maximizes validation micro-F_1 over $[0.01, 0.99]$, with higher precision and then higher τ as tie-breakers. At an intersection over union (IoU) of 0.5, predictions are greedily matched one-to-one within class. Protected testing

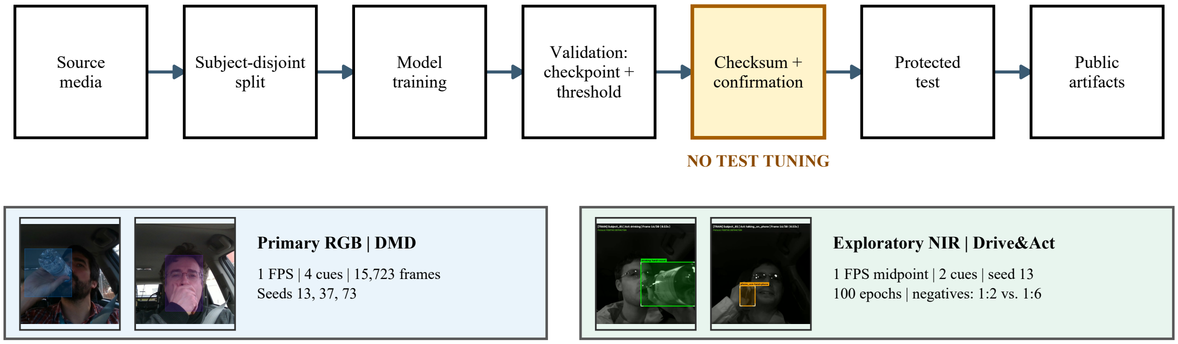

Fig. 1. End-to-end benchmark workflow with representative annotated DMD and Drive&Act frames. RGB and NIR share the validation-before-test gate but retain dataset-specific sampling, ontology, seeds, and training conditions. FPS denotes frames per second.

reports COCO mAP_{50} and $\mathrm{mAP}_{50:95}$ [24], precision, recall, micro- and macro-F_1, and

$$\mathrm{FD}_{100} = 100\frac{\mathrm{FP}_{\mathrm{neg}}}{N_{\mathrm{neg}}}, \tag{1}$$

where $\mathrm{FP}_{\mathrm{neg}}$ is the number of above-threshold false-positive detections on negative frames. We introduce FD_{100} as a simple, interpretable diagnostic: it counts how often the detector fires on frames where no target cue is present. It is a static per-frame count, not a modeled nuisance-alert rate; no temporal confirmation or debouncing is applied.

RGB results are the sample mean and sample standard deviation (SD) over three seeds. Because the subject split is fixed, SD measures optimization variability, not generalization uncertainty across held-out drivers. Each test subject is therefore reported separately, and paired per-seed differences are checked for sign consistency. No significance test is claimed for $n = 3$.

The six YOLO passes were completed before D-FINE-N was added. D-FINE-N selection remained validation-only and its three test manifests were frozen before evaluation; however, because the D-FINE runs were executed in a separate automation suite, the YOLO and D-FINE results are presented together only via the shared test partition, not via a single joint run artifact.

Batch-1 16-bit floating-point (FP16) profiling uses an NVIDIA GeForce RTX 4060 8-GB graphics processing unit (GPU). CUDA synchronization brackets model-forward and tensor-to-final-detection intervals. Disk input/output, decoding, preprocessing, and metric computation are excluded. We report p50 tensor-to-final latency and sustained FPS; these measurements do not represent central-processing-unit or embedded-device deployment.

E. Exploratory NIR Protocol

The secondary NIR source pool contains 3,786 one-second front-top active-NIR (850 nm) Drive&Act snippets from 15 drivers. Twenty-three unilluminated task snippets are excluded before splitting, leaving 3,763. Drinking and phone-use tracklets provide two positive classes. Each snippet contributes one deterministic frame at 1 FPS: the temporal midpoint at $t = 0.5$ s, or zero-based source-frame offset 14. The corresponding box is linearly interpolated at the same instant, then the driver crop is resized to 640×640. Unlike RGB, NIR therefore has dataset-specific frame exclusion and interpolated annotations.

A fixed 9:3:3 subject partition is shared across all sixteen configured model–condition runs. The eight systems span one-stage convolutional, SSD, and transformer-based detector families. Validation and test contain identical ordered examples. Training always contains 270 positive snippets and either 540 or 1,620 negative snippets, corresponding to nominal positive:negative ratios of 1:2 and 1:6. The 1:2 negatives are a subject-stratified seed-13 subset of the 1:6 pool.

All runs use physical batch 8, four-step accumulation, a fixed 100-epoch maximum without early stopping, and the best model-native validation checkpoint. Native optimizers and schedules are retained. The D-FINE augmentation transition is placed at epoch 67. The 1:2 and 1:6 conditions produce 2,600 and 6,000 optimizer updates, respectively. Thus, 1:6 has three times the unique negatives and approximately 2.31 times the optimization updates. It is consequently a *training-negative exposure study*, not a causal class-ratio ablation under matched training signal.

All NIR jobs use seed 13, and operating metrics follow the RGB protocol. Since one frame represents each snippet, no temporal aggregation is performed. Micro-F_1 and FD_{100} retain the detection-level definitions above; macro-F_1 uses the stored class-presence localization metric, in which repeated same-class detections on a frame count once.

IV. Results and Analysis

A. Aggregate RGB Results

Table III summarizes all nine RGB runs. YOLO11n leads mAP_{50} and throughput, YOLO26n leads $\mathrm{mAP}_{50:95}$, and D-FINE-N leads micro- and macro-F_1. The D-FINE advantage is concentrated at the calibrated operating point rather than strict-IoU AP and costs 8.0–26.4 FPS relative to the YOLO systems. The evidence supports an accuracy–efficiency trade-off, not universal superiority.

TABLE III

MEAN AND SAMPLE STANDARD DEVIATION ACROSS THREE SEEDS FOR THE RGB TRACK ON THE PROTECTED TEST SPLIT. BOLD INDICATES THE BEST RESULT PER COLUMN.

Model	mAP_{50}	$\mathrm{mAP}_{50:95}$	Micro-F_1	Macro-F_1	FD_{100}	p50 (ms)	FPS	Params (M)	GFLOPs
YOLO11n	**91.79±0.44**	55.38±0.41	86.60±0.49	83.42±1.36	**0.63±0.04**	**14.30±0.20**	**68.0±1.3**	2.59	6.44
YOLO26n	89.28±0.99	**56.98±0.81**	84.44±1.14	79.39±1.49	0.67±0.06	19.21±0.09	51.7±0.2	**2.51**	**5.78**
D-FINE-N	90.22±2.29	55.38±0.89	**91.14±2.66**	**84.33±4.51**	0.87±0.21	23.31±0.46	40.4±1.2	3.72	7.44

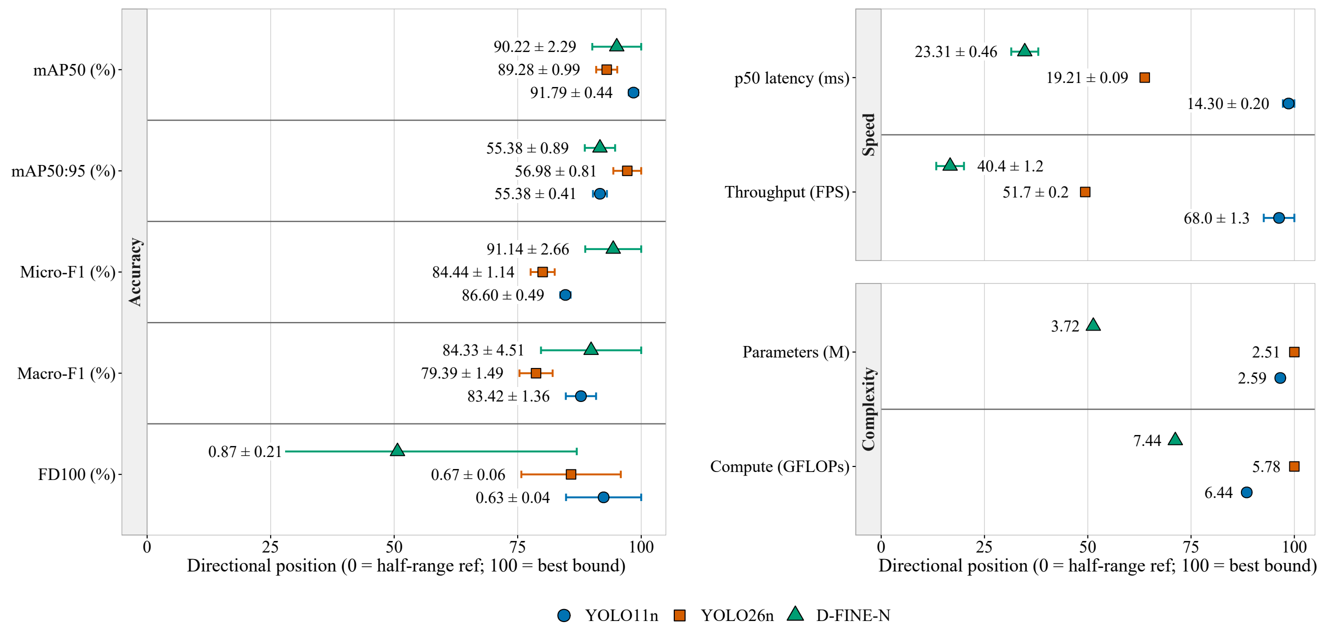

Fig. 2. Directional dot-and-whisker plot for YOLO11n, YOLO26n, and D-FINE-N. Rows are grouped into accuracy (left), speed, and complexity (right); right indicates better performance, and whiskers show sample SD across seeds.

B. Class Imbalance and Paired-Seed Evidence

Phone use accounts for 497 of 614 positive test instances, whereas hand over mouth, yawning, and drinking have only 28, 33, and 56 instances. This imbalance can influence learned representations, validation-selected thresholds, and the pooled micro-F_1. It does not directly dominate COCO mAP, because AP is first computed by category and then averaged equally across the four categories.

The class-wise mean differences, YOLO26n minus YOLO11n, are $+6.21$ points for hand over mouth, -3.85 for yawning, $+1.02$ for drinking, and $+3.02$ for phone use. D-FINE-N obtains the highest mean AP for yawning, drinking, and phone use, but the lowest for hand over mouth. Reporting macro-F_1 and per-class AP beside micro-F_1 prevents the majority phone-use class from silently defining the conclusion, although the wide SD for low-support cues still warrants caution.

C. Held-Out Subject Sensitivity

Averaged across seeds, YOLO11n micro-F_1 is 94.94%, 62.41%, and 94.72% for Subjects 05, 10, and 12; YOLO26n obtains 96.11%, 54.11%, and 92.29%; and D-FINE-N obtains 98.20%, 78.79%, and 94.30%. The between-subject change is far larger than the pooled seed SD. Reporting only the pooled test set would therefore hide the difficult driver.

D. Qualitative Error Analysis

Figure 3 shows predeclared seed-13 examples from frozen predictions for all three RGB detectors. Correct detections localize the intended interaction, while the false-negative examples expose glare and small drinking targets. More importantly, nominal negatives can appear visually cue-like. They may reflect difficult ontology boundaries or missed labels, so the structural annotation audit alone cannot establish semantic validity. The examples are retained because they expose the benchmark's principal annotation risk.

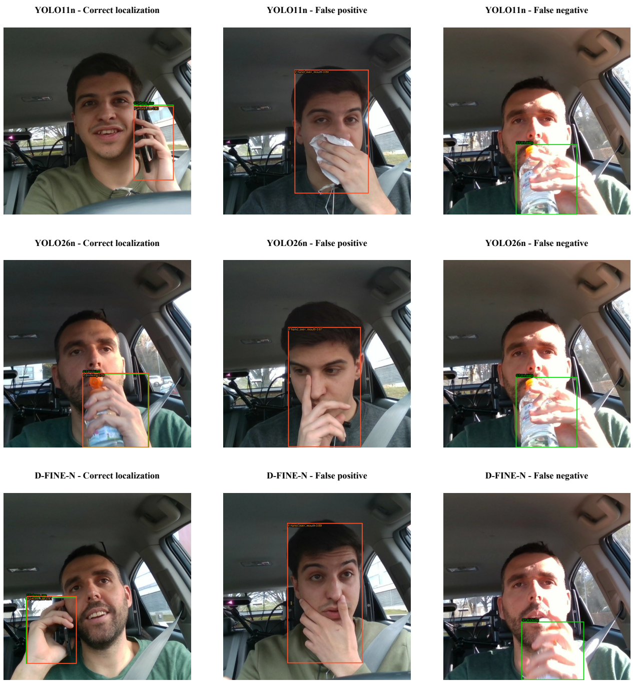

Fig. 3. Representative seed-13 outcomes for YOLO11n, YOLO26n, and D-FINE-N. Green boxes denote ground truth and orange boxes denote predictions at the validation-selected threshold; columns show a correct localization, a false detection on a nominal negative, and a false negative.

E. Exploratory NIR Results

All 16 protected NIR passes are complete. Table IV reports both exposures for YOLO11n, YOLO26n, D-FINE-N, SSDLite-MobileNetV3-Large, RT-DETRv2-S, YOLOX-Nano, YOLOv10n, and YOLOv8n, while Fig. 4 plots the same checksum-backed results.

Increasing negative exposure from 1:2 to 1:6 improves $\mathrm{mAP}_{50:95}$ for only YOLO26n, SSDLite-MobileNetV3-Large, and RT-DETRv2-S; it lowers AP for the other five systems. The changes range from $+5.52$ percentage points for YOLO26n to -7.71 for YOLOX-Nano. Micro-F_1 improves only for SSDLite-MobileNetV3-Large and decreases for the other seven systems. The direction is therefore model-dependent rather than a uniform benefit of 1:6 exposure. RT-DETRv2-S has the highest strict-IoU AP at both ratios, while YOLOv8n has the highest micro-F_1 and lowest tensor-to-final p50 latency at both ratios.

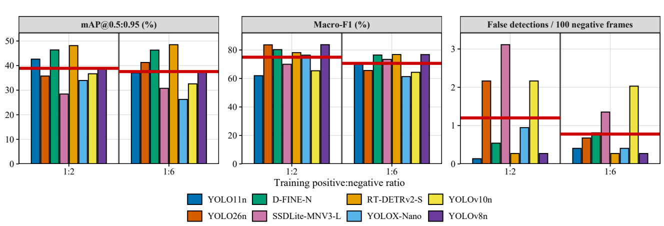

Fig. 4. Sixteen NIR protected-test results at seed 13 and deterministic 1-FPS midpoint sampling. Grouped columns compare all eight models at 1:2 and 1:6 training exposures; the dark-grey rule separates the ratio sections, and each thick red segment marks the eight-model mean for its ratio.

TABLE IV
EXPLORATORY NIR TEST RESULTS AT SEED 13

Model	Neg.	$\mathrm{mAP}_{50:95}$	Micro-F_1	Macro-F_1	FD_{100}	p50
YOLO11n	1:2	42.62	71.91	61.94	0.14	15.22
YOLO11n	1:6	37.07	69.15	70.32	0.41	13.74
YOLO26n	1:2	35.75	76.54	83.62	2.17	17.11
YOLO26n	1:6	41.28	65.96	65.57	0.68	16.84
D-FINE-N	1:2	46.38	83.58	80.25	0.54	24.58
D-FINE-N	1:6	46.30	79.23	76.45	0.81	24.10
SSDLite-MNV3-L	1:2	28.43	73.39	69.97	3.11	14.76
SSDLite-MNV3-L	1:6	30.76	78.73	73.38	1.35	15.93
RT-DETRv2-S	1:2	48.14	84.42	78.14	0.27	25.61
RT-DETRv2-S	1:6	48.51	79.79	76.80	0.27	24.66
YOLOX-Nano	1:2	33.95	79.23	76.36	0.95	15.46
YOLOX-Nano	1:6	26.24	67.37	61.31	0.41	16.24
YOLOv10n	1:2	36.67	69.23	65.40	2.17	15.41
YOLOv10n	1:6	32.58	66.67	64.32	2.03	13.45
YOLOv8n	1:2	39.21	84.51	83.71	0.27	11.74
YOLOv8n	1:6	37.77	80.81	76.75	0.27	10.24

V. DISCUSSION

A. Accuracy–Efficiency Trade-offs

No detector is universally best: YOLO11n leads coarse-IoU mAP and throughput, YOLO26n leads strict-IoU mAP and compactness, and D-FINE-N leads micro- and macro-F_1 but runs slowest in RGB. In NIR, RT-DETRv2-S leads strict-IoU AP, whereas YOLOv8n leads micro-F_1 and is fastest at both exposures. Native training and postprocessing make these shipped-system comparisons, not isolations of architecture.

B. Class and Subject Effects

The phone-use majority is a real benchmark characteristic and a limitation for pooled operating metrics, but it does not invalidate YOLO26n's strict-IoU advantage. Equal-weight class AP, macro-F_1, per-class supports, and paired-seed directions separate that result from majority-class dominance. The narrow aggregate margin, low support for three categories, and drinking sign reversal mean the accurate conclusion is a trade-off rather than a definitive rank. The difficult Subject 10 further shows that seed variation on a single split understates deployment variation across people.

C. Interpretation Scope

Subject-disjoint evaluation reduces identity leakage but does not establish population-level generalization from only 14 drivers and three held-out test identities. A single annotator and one pass provide internally consistent data but no independent semantic agreement estimate; the qualitative nominal-negative case makes this concrete. Correlated frames remain within RGB recordings, and single-frame detection neither estimates temporal fatigue nor models alert debouncing. NIR avoids within-snippet duplication through one midpoint frame, but 1-FPS sampling does not characterize motion within the second. NIR additionally differs in dataset, ontology, annotation source, and preprocessing, uses one seed, and changes negative diversity together with training steps. It cannot be interpreted as a controlled RGB-versus-NIR modality effect. Finally, RTX 4060 timing excludes preprocessing and is not an embedded-hardware measurement. These boundaries define what the benchmark supports without removing its value as a transparent, reproducible comparison.

VI. Conclusion and Future Work

The RGB comparison establishes a system-level trade-off: YOLO11n leads coarse-IoU accuracy and throughput, YOLO26n leads strict-IoU AP and compactness, and D-FINE-N leads pooled and class-balanced F_1 but is slowest. Class, subject, and suite-extension evidence prevent a universal winner or jointly frozen nine-run claim. The 16 completed NIR passes show no uniform strict-IoU benefit from 1:6 exposure; micro-F_1 improves only for SSDLite-MobileNetV3-Large, RT-DETRv2-S leads strict-IoU AP, and YOLOv8n leads micro-F_1 at both ratios.

Future work will add independent multi-annotator review, increase rare-cue and held-out-driver coverage, repeat the benchmark across subject partitions and external cabins, and test class-balanced or matched-step training. Temporal alert logic will connect frame localization to user-facing warnings. The NIR study will be strengthened with matched optimization exposure and additional seeds, while embedded-device profiling will include preprocessing and end-to-end input latency. These extensions directly target the uncertainty exposed by the present class, subject, and qualitative analyses.

Data and code: Authored annotations, frozen partitions and checksums, sanitized predictions, evaluation tools, and publication artifacts are available at https://github.com/IamOumarIbrahim/lightweight-driver-behavior-detection; licensed DMD and Drive&Act media and trained checkpoints remain local under their original access terms.

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